



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

M. Kumarar
192525022

ITA0506- COMPUTER VISION

MODEL LAB PRACTICALS

SET 7

1. Implement the Erosion Morphological operations technique using Open CV in python.
2. Implement the Dilation technique as a Morphological operation to dilate the foreground regions based on Open CV.
3. Implement the Opening technique as a Morphological operation to dilate the foreground regions based on Open CV.
4. Write a Python function to extract the text from videos.

Rivaattunale

88/100

Model Practical

computer vision

Name: M. Kumaran

Reg. No: 192525022

Course Code: ITA0506

Date: 03/09/2026

1) Aim:

→ To implement the Erosion Morphological operations technique using Open CV in Python.

Algorithm:

- (i) Read the Input Image.
- (ii) Convert the image into grayscale.
- (iii) Apply Binary Thresholding.
- (iv) Create a morphological kernel.
- (v) Apply cv2.erode().
- (vi) Display the original and eroded image.

code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

image = cv2.imread("input.jpg").
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```


cv2.cvtColor

$\rightarrow$ binary = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)

kernel = np.ones(5, 5), np.uint8)

plt.show()

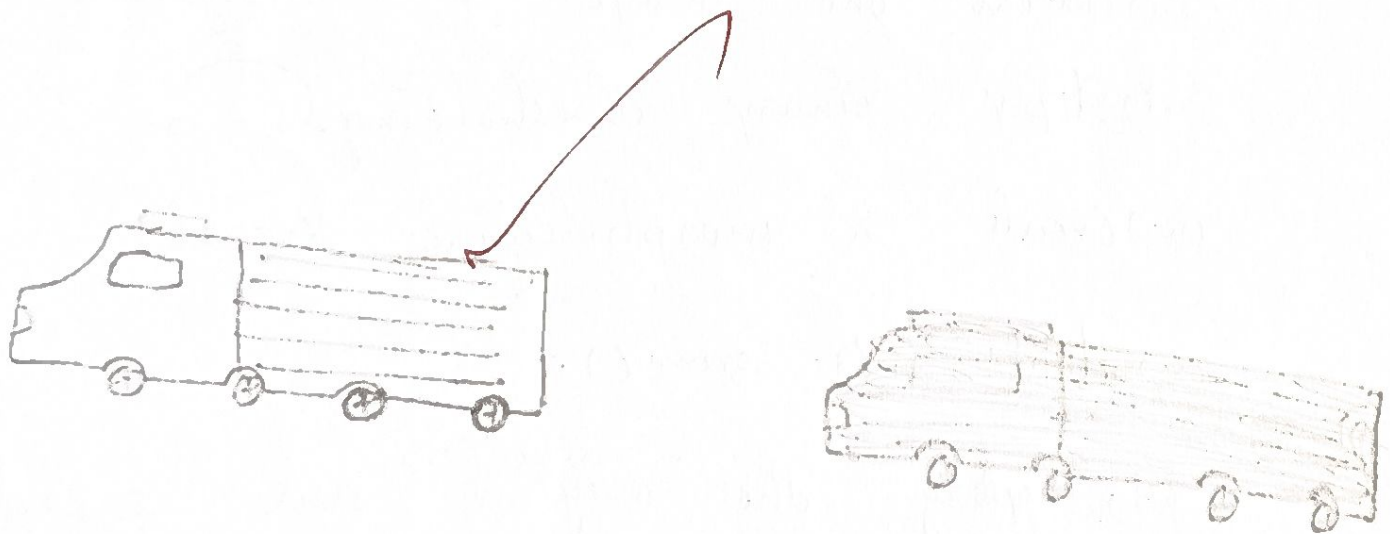
Input:

A binary image containing foreground objects.

Output:

The foreground objects become smaller and thinner after erosion.

Result:



Result:

Therefore, the technique is executed using open cv.

Aim: To implement the Dilation technique as a Morphological operation to dilate the foreground regions based on open cv.

Algorithm:

Read the Input Image.

convert it to grayscale.

Apply binary thresholding.

Create a kernel.

Apply cv2.dilate().

Display the original and dilated Images.

code:

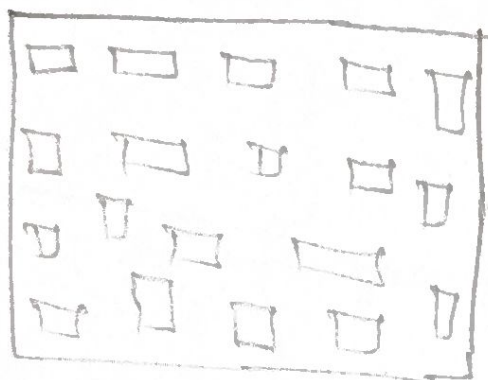
```
import cv2
import numpy as np
img = cv2.imread("input.jpg", 0).
kernel = np.ones(5, 5), np.uint8)
result = cv2.dilate(img, kernel, iterations=1).
cv2.imshow("Dilation", result).
cv2.waitKey(0).
cv2.destroyAllWindows()
```


Input:

A binary image containing foreground regions.

Output:

The foreground objects become larger & thicker.



Result:

Hence, the Dilation technique as a Morphological operation using Open CV executed.

Aim:

→ To implement the opening morphological operation for processing based on Open CV.

Algorithm:

Read the input image
Convert it into grayscale.
Apply Binary thresholding.
Create a morphological kernel.
Apply erosion followed by dilation.
Display the result.

code:

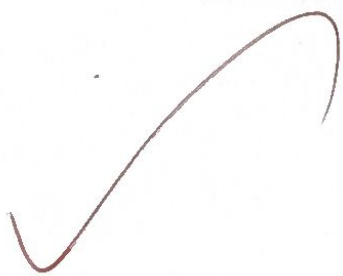
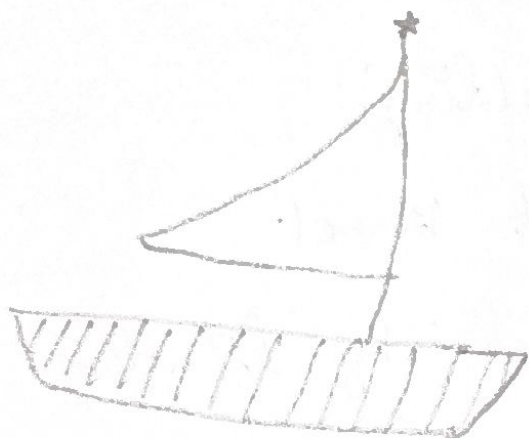
```
import cv2
import numpy as np
img = cv2.imread("input.jpg")
kernel = np.ones(5,5), np.uint8)
result = cv2.morphologyEx(img, cv2.MORPH_OPEN,
                           kernel)
cv2.imshow("opening", result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```


Input:

A Binary Image containing objects and small noise.

Output:

Small unwanted noise is removed while the major foreground objects are retained.



Result:

Therefore, the technique is executed by open cv.

aim:

To write a Python function for extracting text from video frames using opencv and ocr.

Algorithm:

Open the video file

Reach each frame.

Convert the frame to grayscale.

Apply thresholding.

Perform OCR on the processed frame.

Store the Extracted Text.

Continue until all frames are processed

Display the extracted text.

Code:

```
import cv2.
```

```
import pytesseract.
```

```
cap = cv2.VideoCapture("Input.mp4")
```

```
while True:
```

```
    ret, frame = cap.read()
```

```
    if not ret:
```

```
        break
```

```
    text = pytesseract.image_to_string(frame)
```


cap. release ()

Input:

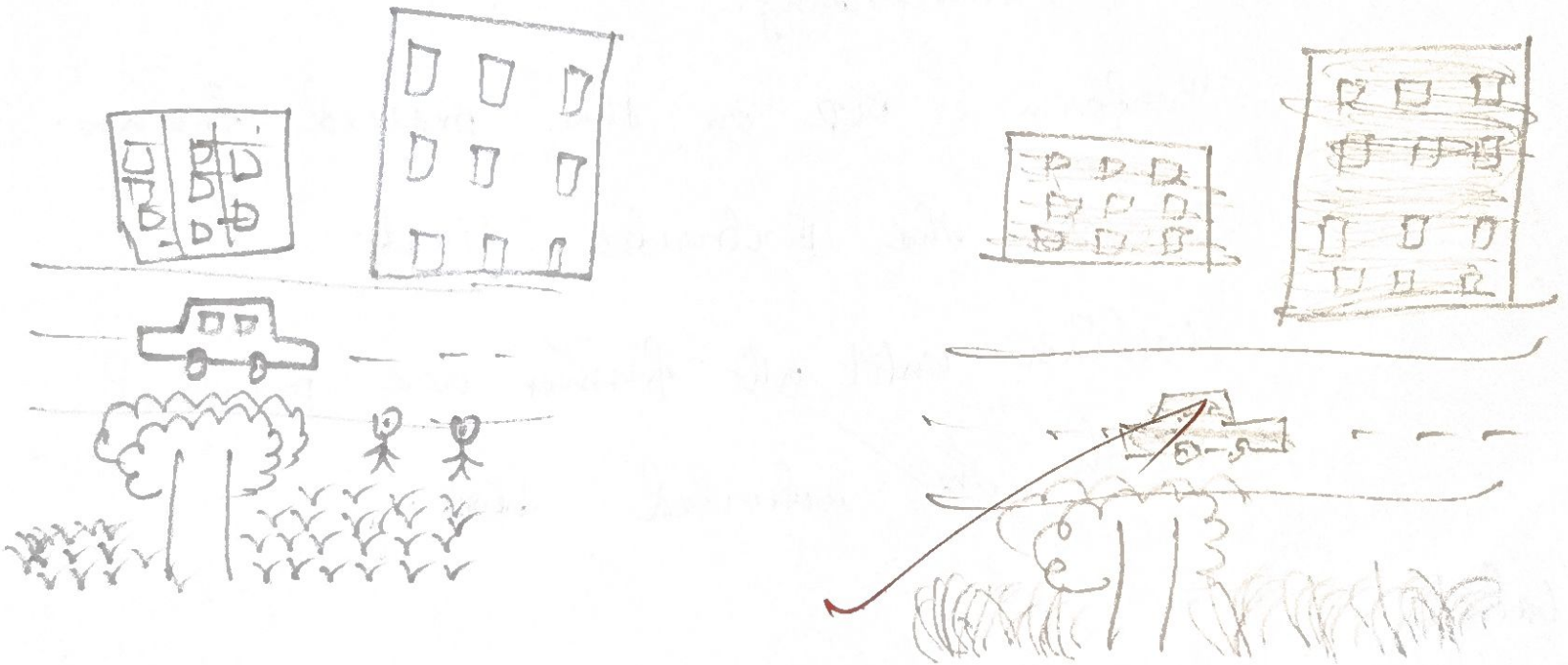
A video containing visible text, for

example :

input - video . mp4.

Output:

The Program extracts the recognized text from the video frames and displays it.



Result:

Therefore, the difference of the video is seen.